

NCL Snowflake ML Modernization - Proposed MLOps Foundation

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Blend360 case study brief | Norwegian Cruise Line Snowflake ML modernization proposal

Executive Summary

Norwegian Cruise Line (NCL) sought to modernize a portfolio of approximately 20 regression-type marketing and sales models, primarily built and deployed in R. The models were not yet designed for scalable production practices such as CI/CD; some were deployed through Salesforce Einstein, while others ran manually on VMs.

Blend360 proposed a six-month, retainer-based cross-functional pod to migrate high-value models into Snowflake ML, establish Git-driven MLOps and CI/CD workflows, build a governed Snowflake-native feature store, and implement monitoring, governance, and enablement. The materials describe a proposed delivery plan; no NCL implementation results are provided.

The Challenge

NCL's model portfolio relied primarily on R and had fragmented deployment patterns across Salesforce Einstein and manual VM processes. Model scoring tables were batch-processed and stored in Snowflake, with activation through internal tools including Tableau and Salesforce and exploratory use of Streamlit. NCL identified a need to standardize ML infrastructure, move development into a Git-based MLOps environment, consolidate features into a governed feature store, and define model testing, monitoring, retraining, and post-deployment success measures.

Our Solution

Blend360 proposed an infrastructure/readiness audit followed by migration of high-value R models to production-grade Python/Snowflake ML pipelines. The proposed solution included Git-based workflows and CI/CD, automated retraining and scoring pipelines, model versioning and release processes, a Snowflake-native feature store, and monitoring dashboards in Tableau and/or Streamlit for model health, data drift, and performance. The proposed delivery model was a dual-skilled retainer pod of data engineering and data science practitioners, with MLOps practice leadership included as a Blend investment.

Key Features

- Infrastructure Readiness Audit - Review current models, pipelines, data sources, Snowflake, and downstream-tool connections to identify build and deployment risks or gaps.
- R-to-Snowflake ML Migration - Convert high-value R models into production-grade pipelines using Python and Snowflake ML.
- Git-Driven MLOps and CI/CD - Establish versioning, automated retraining, structured releases, and orchestration for model scoring pipelines.
- Snowflake-Native Feature Store - Design a shared, governed feature store with data science-led feature selection and engineering to support reuse across the data science team.
- Monitoring, Governance, and Enablement - Implement model-health, drift, and performance monitoring; document deployment, testing, retraining, and impact-measurement playbooks; and transfer framework ownership to NCL.

Quantified Outcomes

- Known gap: the provided materials do not include quantified outcomes for an implemented NCL engagement.

Business Value

This is a credible modernization capability story for organizations with business-critical models that are valuable but difficult to scale, govern, and operate consistently. The proposed approach connects legacy-model migration with reusable feature management, production controls, monitoring, and internal-team enablement—creating a repeatable path from isolated model operations toward enterprise ML lifecycle management. The brief should be positioned as a proposal and delivery approach, not as evidence of completed NCL outcomes.

Known Gaps / Caveats

- Implementation status and completed scope are missing. Ask the NCL account lead or delivery lead whether the proposal was sold, started, or completed; this would determine whether the brief can be presented as a delivered case study rather than a proposed capability story.
- NCL-specific outcomes, baseline metrics, and realized business impact are missing. Ask the NCL sponsor and project owner for deployment speed, model reliability, model-performance, adoption, cost, revenue, or efficiency measures; this would substantiate the value proposition with client-specific evidence.
- Final model inventory, migration count, and prioritized use cases are missing. Ask the NCL data science lead for the confirmed inventory and delivery backlog; this would clarify the scale and business relevance of the work.
- Final technical architecture and tools selected beyond Snowflake ML, Git-based workflows, Tableau, and exploratory Streamlit usage are missing. Ask the technical architect or delivery lead for the approved architecture; this would make the technology story precise.
- The proposal contains indicative six-month timelines and commercial options, but no confirmed statement of work or final commercial structure. Ask the account lead or procurement owner for the executed scope; this would prevent proposed milestones or pricing from being represented as delivered commitments.

Sources Used

- 25.7.14_NCL_Proposal_V1 - Copy.pptx, slides 2-3: NCL model landscape, modernization needs, proposed opportunity, and key deliverables.
- 25.7.14_NCL_Proposal_V1 - Copy.pptx, slides 4-5: illustrative six-month delivery timeline, proposed MLOps activities, enablement, and team responsibilities.
- 25.7.14_NCL_Proposal_V1 - Copy.pptx, slides 5-6: proposed retainer delivery options, indicative milestones, and Blend investments.